{EPITECH}

HOMEPEDIA_

< BOOTSTRAP />

HOMEPEDIA

This bootstrap will guide you through 4 main steps of the project:

- ✓ data gathering ;
- ✓ database organization ;
- ✓ big data analysis ;
- ✓ interactive visualizations.

All datasets needed for this bootstrap are available on the Sharepoint site T-DAT-902.

Non tabular data gathering

Beside collection of basic csv files, you might also need to do some scraping to extract your data (from websites like pap.fr or leboncoin.fr).

As a useful start, you can try to extract and structure data located on ville-ideale.fr, a website where people can review the city they live(d) in. Try to extract data from as many cities as possible.

Create a dataset to store everything you collected using tools adapted for non tabular data.

> Start with using some official metadata to get information about the cities, their name and link them to the scrapped data

Database organization with SQL

Since your data is coming for various sources (format: csv, txt... origin: Gouvernement, private website), it's important that you clean and organise your data by creating common indexes.

For this exercise you are given a dataset ,named `KaDo.csv`, of sales in a beauty shop. One line is equivalent to one product bought by one client at a specific time. You are also given information such as the different categories the product bought belon to, the name of the product, client and ticket ID, month of purchase and of course price.

With this dataset of seven million rows try to build a relational database with different files for the clients, the products...

> For this relational database exercice you can use tools such as MySQL.

Big data analysis: an introduction with pyspark

A good first step to experiment computing with big data is pyspark.

You can import datasets, display results, grouping and exporting. You can also make SQL queries. Start by creating a SQL database and use pyspark to gather some relevant information about your dataset. Try to know more about different functions of pyspark such as `map()`, `sortByKey()`, `reduceByKey()`.

For the exercise you can use the dataset named `france_total_real_estate_sales_2022.csv.zip`.

Interactive visualisation: a glimpse with Gapminder

You are asked for the project to develop your own visualisation app. Depending on the graphic you produce you will have to structure your data accordingly.

For this step you can find inspiration using the tools from **Gapminder** which allow you to display interactive visualisation. Start by downloading the desktop version of Gapminder.

For this exercise you can work with the dataset named `speech_time_mw.csv` which contains records about the speech time of men and women at french radio and television.

Do some appropriate cleaning and load the data in Gapminder to display vairous information, such as the distribution of men/woman speech time over years on public and private channels.

Gapminder may look in term of visualisations, but it will give you some hint on how to build a well-designed app.

Once you got some ideas, you can start playing with **streamlit** to create your own interactive dashboard as you will have to do for the project.

The choice of the tool is at your discretion.
A good one could be plotly.

Cartographic visualisation

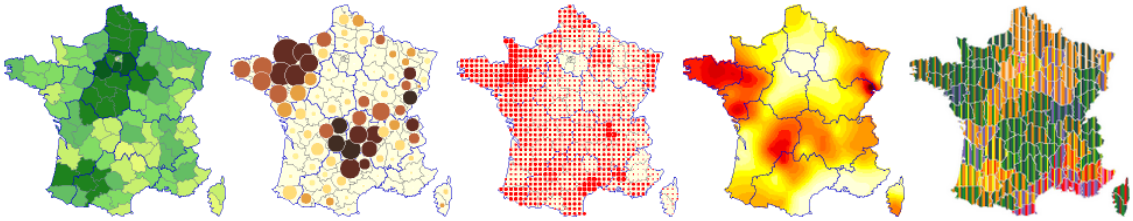
When making analysis related to a territory, cartographic visualisation come in mind right away.

To prepare yourself to advance cartographical analysis you can start by downloading the ”pathology” dataset which give your number of patients by pathology over the territory.

Then find on internet a shapefile for the french departments. Then display a heatmap showing the repartition of a specific pathology (which ever).

Try to improve with the colors, legend and titles.

Many libraries can help you, in python look for matplotlib of plotly. If you have Windows, the french software *philcarto* is interesting to play with.



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